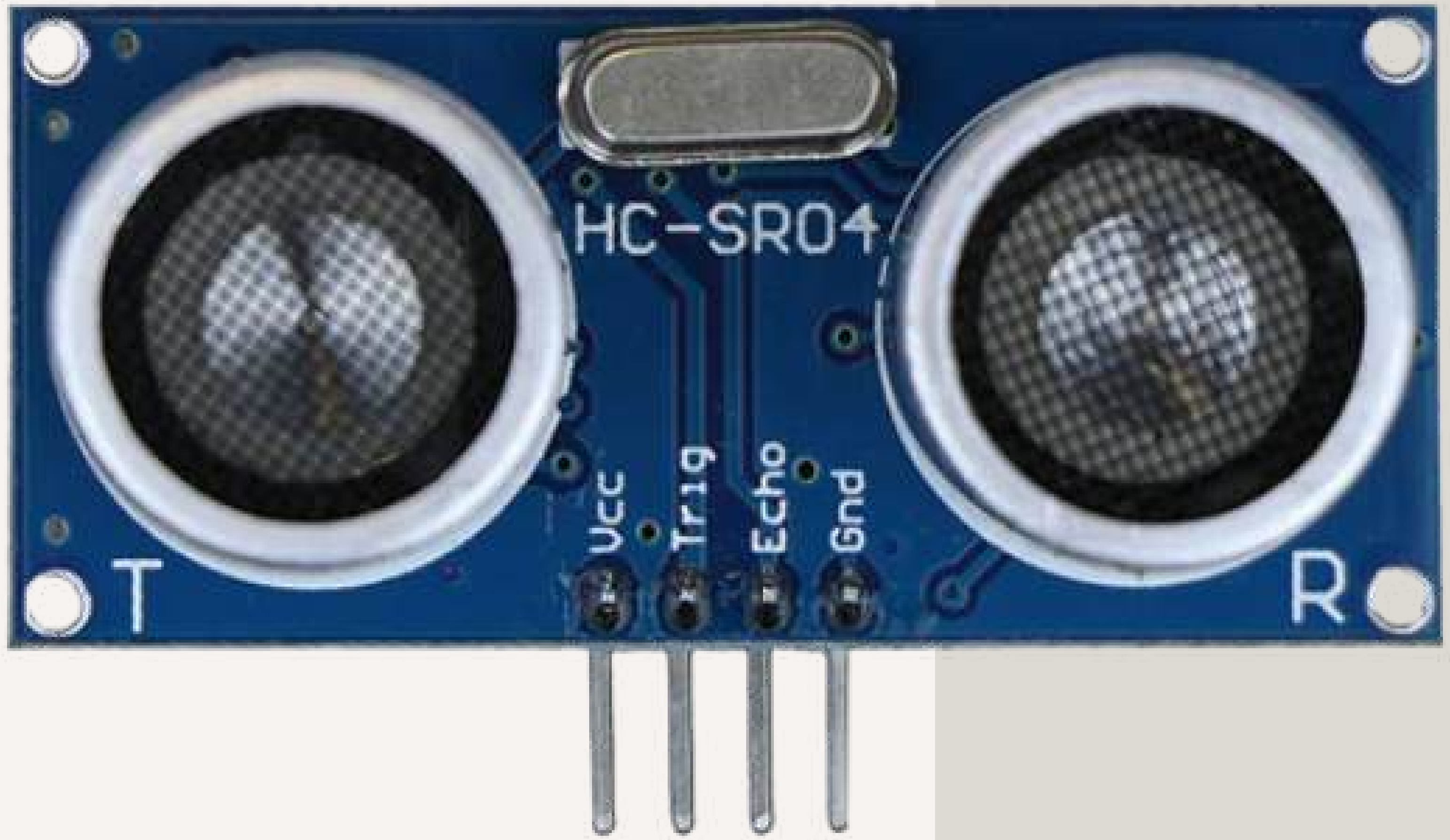
Lava Academy

12TH MAY 2026

ULTRASONIC **SENSOR**

Distance Sensing through Echolocation and Sound Waves



POWERED BY

DIRECTOR OF RESEARCH
MECHATRONICS ENGINEERING

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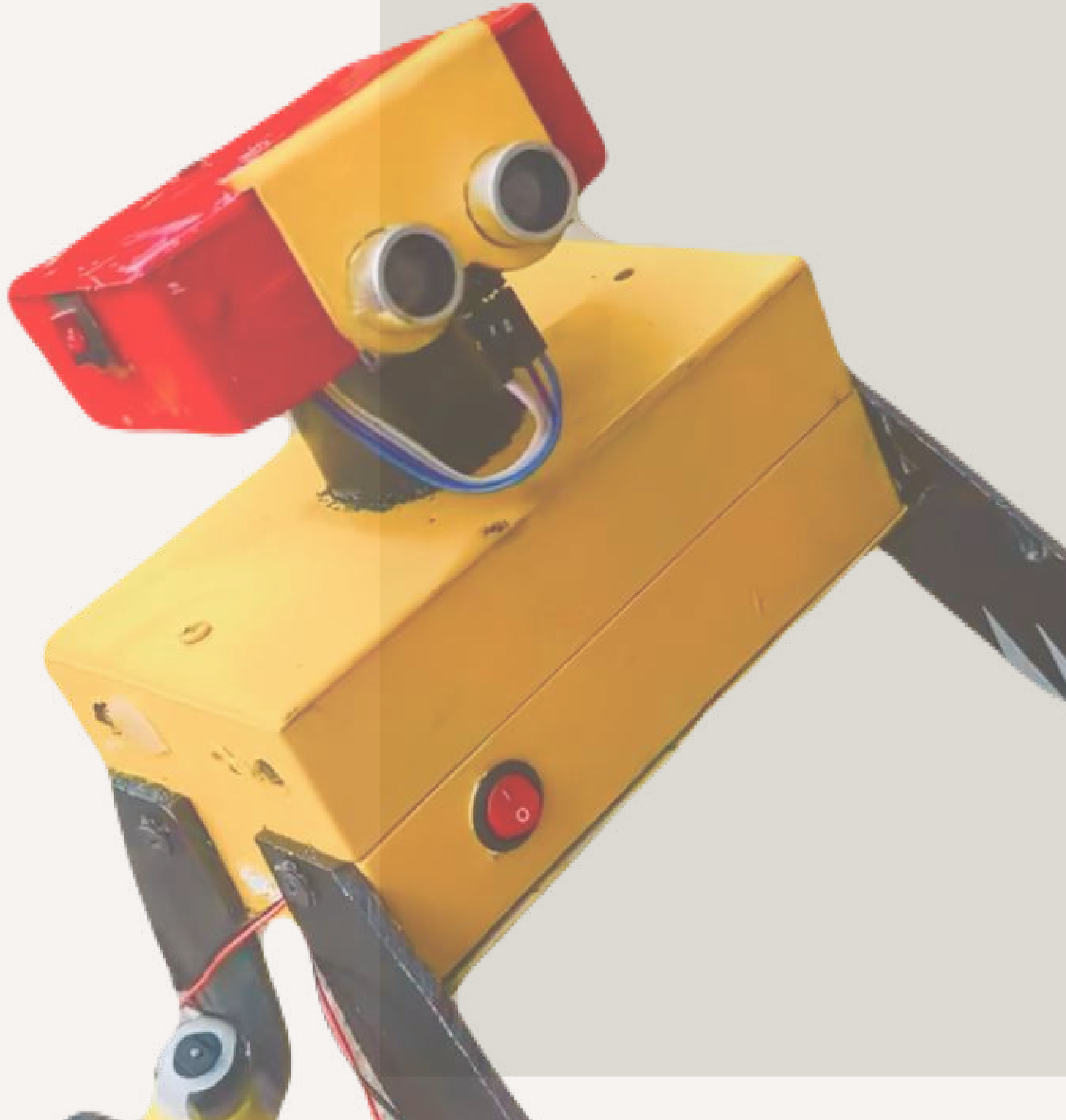
WHAT IS ULTRASONIC SENSOR?

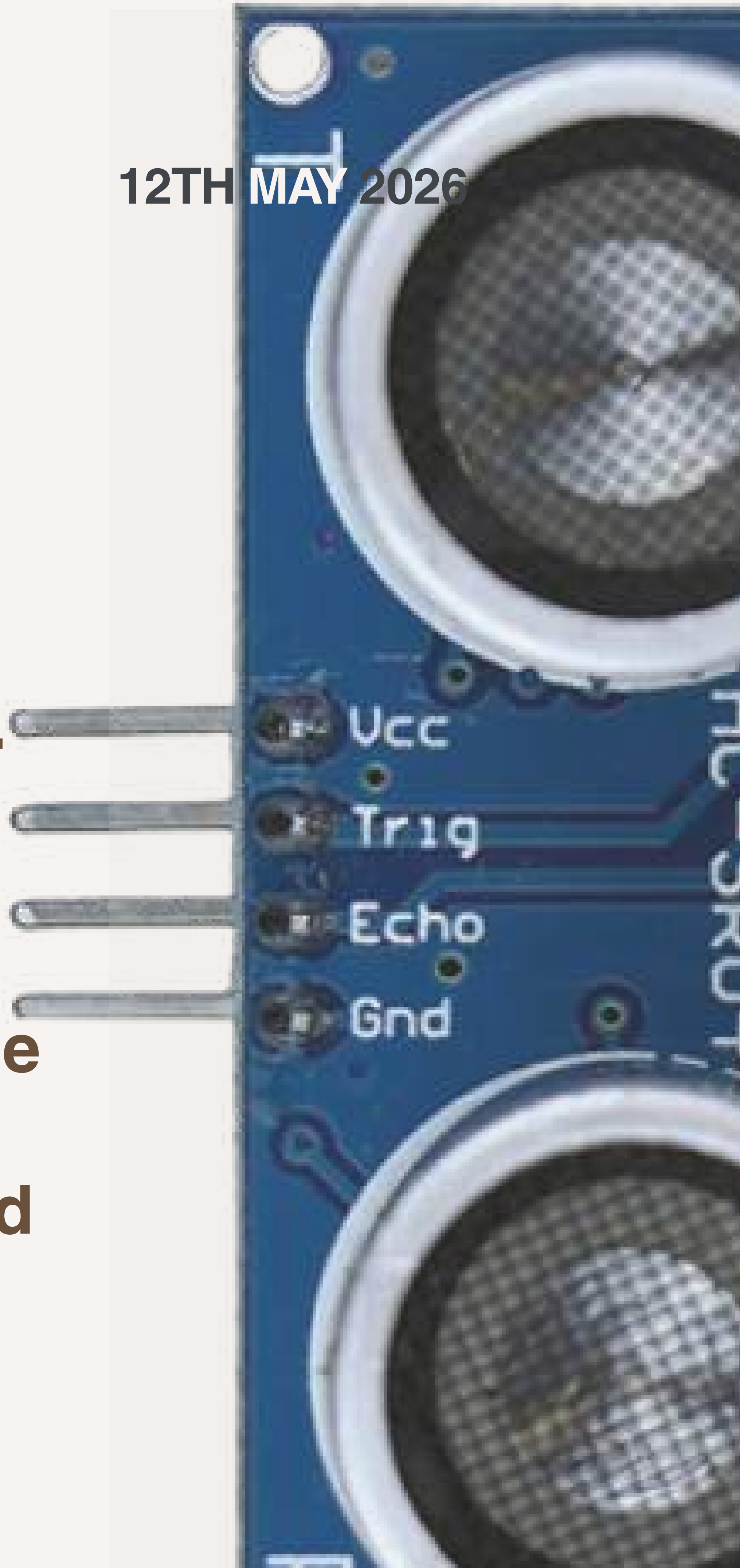
An ultrasonic sensor is an electronic device that measures distance by emitting high-frequency sound waves and timing how long it takes for the echo to bounce back from an object. It mimics the natural echolocation used by bats, operating at a frequency of 40kHz which is well above the range of human hearing.



SENSOR ANATOMY & PINOUT

- **The Hardware:**
- **Transmitter (T): The "mouth" it emits 40kHz sound bursts.**
- **Receiver (R): The "ear" it listens for the returning echo.**
- **The 4 Pins:**

1. **VCC: Power (5V DC).**
2. **Trig (Trigger): Input pin. Send a 10µs pulse here to start the measurement.**
3. **Echo: Output pin. Stays HIGH for the duration of the sound wave's travel.**
4. **GND: Ground connection.**



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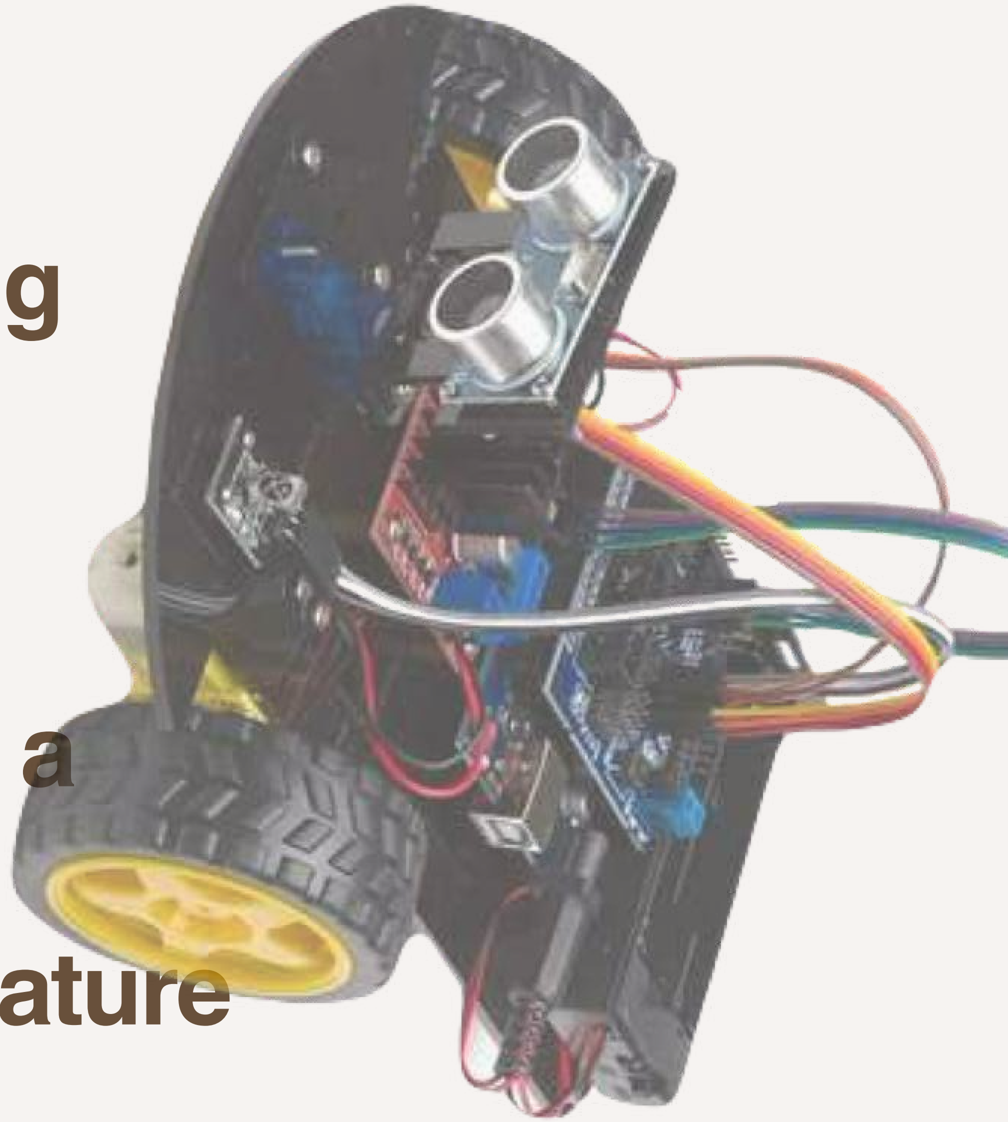
THE PHYSICS OF ULTRASOUND



- Sound Frequency: Operates at 40kHz (Human hearing stops at 20kHz).

The Method: Time-of-Flight (ToF).

- Distance is calculated based on how long it takes for a sound wave to hit an object and return.
- The Environment: Performance is affected by temperature and air density, but for basic robotics, we assume a constant speed of sound

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WORKING PROTOCOL

- Step 1: The microcontroller (Arduino/ESP32) pulls Trig HIGH for 10µs.
- Step 2: The sensor automatically sends out an 8-cycle sonic burst.
- Step 3: The Echo pin goes HIGH immediately.
- Step 4: The sensor waits for the reflected signal. When detected, the Echo pin goes LOW.
- Teaching Point: The width of the Echo pulse is what we measure (using the pulseIn() function).

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THE MATH (CALCULATION)

- The Core Formula:
- Distance = {Time \times Speed\ of\ Sound} divided by{2}

Constants for Code:

- Speed of sound = 343 m/s (or 0.0343 cm/µs).
- The "Divide by 2" Rule:
- Crucial point: The sound travels to the object and back. To get the actual distance to the object, we must halve the total travel time.

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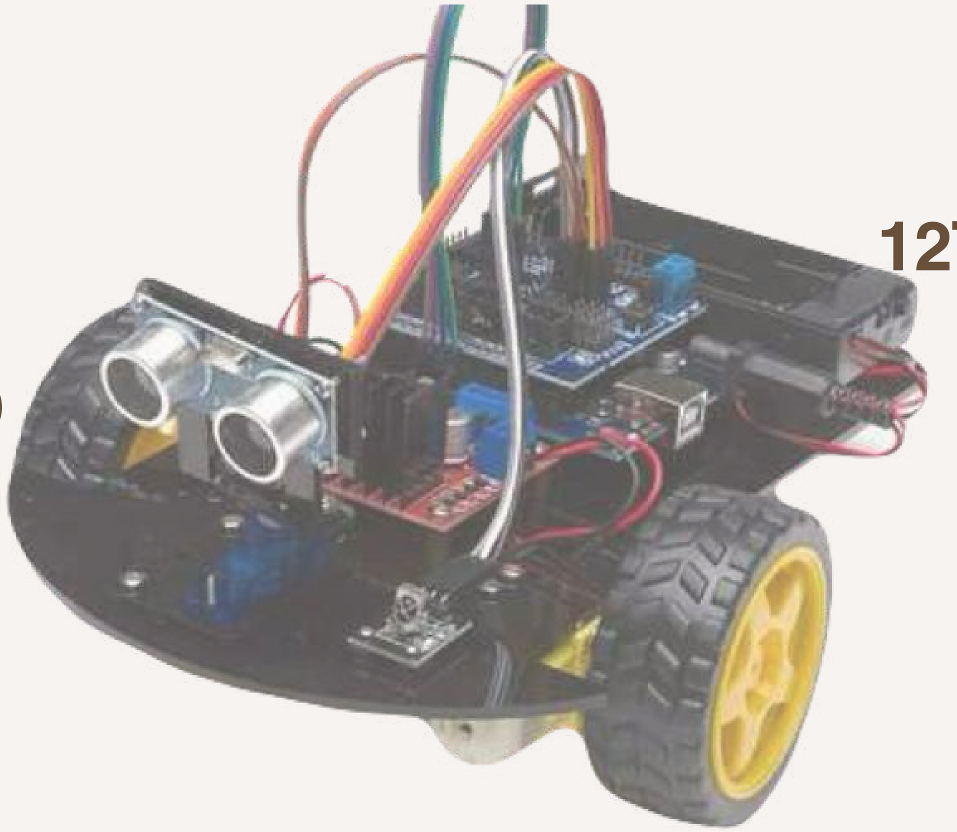
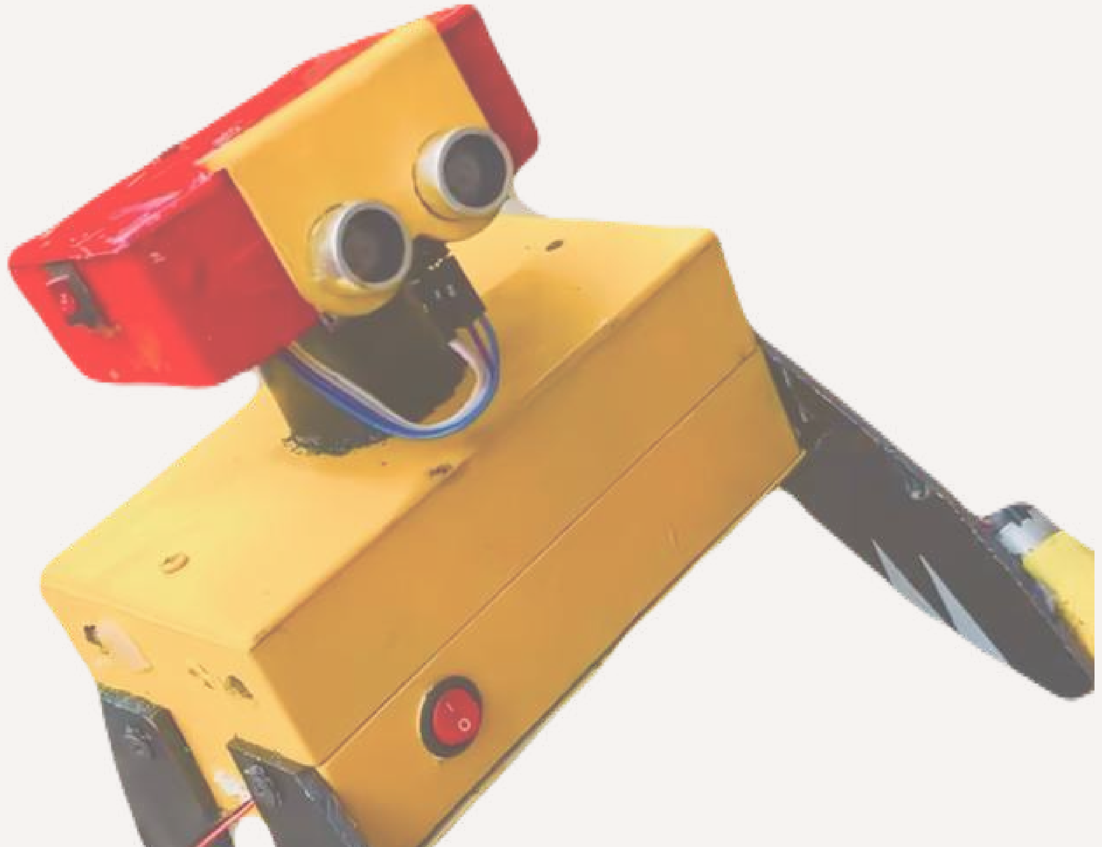
PRACTICAL APPLICATIONS



Obstacle avoidance (Battlebots).

- Liquid level sensing (Water tanks).
- Parking sensors (Automotive).



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LIMITATIONS :

- **Absorption: Soft surfaces (like sponges or thick cloth) soak up the sound.**
- **Deflection: Smooth surfaces at a sharp angle reflect the sound away from the receiver.**
- **Dead Zone: Objects closer than 2cm usually cannot be detected.**

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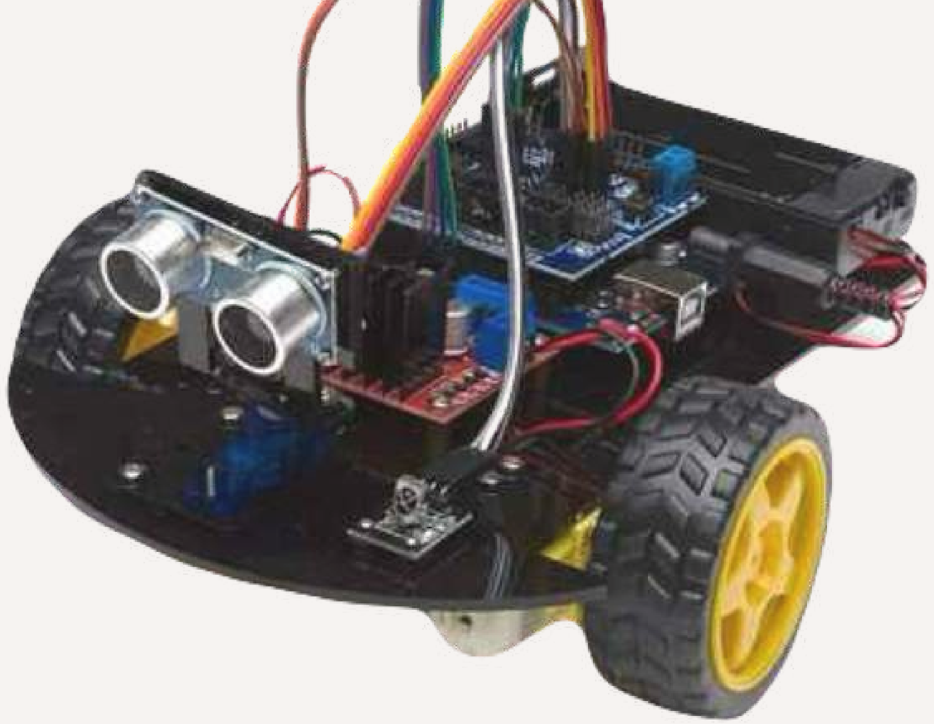

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THANK YOU!

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